The Journal of Open Source Software

LithoGas: Monte Carlo model for estimating H_2 and ^{4}He generation through radiolysis and serpentinization

Daniel S. Coutts[1¶], **Oliver Warr**[2], **Omid H. Ardakani**[1], **and Barbara Sherwood Lollar**[3,4]

1 Natural Resources Canada - Geological Survey of Canada - Calgary **2** University of Ottawa, Canada **3** University of Toronto, Canada **4** Institut de Physique du Globe de Paris (IPGP) ¶ Corresponding author

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor:

Submitted: 10 October 2026
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0)

Summary

Naturally occurring hydrogen (H_2) in the Earth's subsurface represents a novel source of H_2 for use in electricity generation and hard-to-abate industrial processes (e.g., ammonia, fertilizer, steel) (Ballentine et al., 2025; IEA, 2024; Johnson et al., 2025; Sherwood Lollar et al., 2025). Helium (He), meanwhile, is a critical non-renewable resource utilized in medical, industrial, and research fields (Anderson, 2018; Danabalan et al., 2022). With a growing list of H_2 and He showings in different geological settings (Danabalan et al., 2022; Miaga et al., 2023; Sherwood Lollar et al., 2014; Sherwood Lollar & Warr, 2026; Truche et al., 2024) along with focused exploration for these gases [Jackson et al. (2024); Ballentine et al. (2025); Hu & Li (2025); SherwoodLollar:2025], new tools are required to prospect for and model these emerging resources. Natural H_2 is produced through two dominant geologic processes (Ballentine et al., 2025; Sherwood Lollar et al., 2014): radiolysis (Higgins et al., 2025; Lin et al., 2005; Warr et al., 2023) and serpentinization [Coveney et al. (1987); SherwoodLollar:1993; Mayhew et al. (2013)]. Radiolytic H_2 is generated when ionizing α, β, and γ radiation emitted by U, Th, and K-bearing minerals dissociates water into reactive species, predominantly forming molecular hydrogen (H_2) and hydrogen peroxide (H_2O_2). Dissolved oxygen (O_2) may subsequently form via the secondary decomposition of H_2O_2.It is important to note that released alpha particles are equivalent to ^{4}He, and the breakdown of U and Th-bearing minerals is also the dominant process for the generation of helium in the subsurface. H_2 formation through serpentinization meanwhile requires the release of H_2 due to the oxidation state change of Fe^{2+} to Fe^{3+} in different mineral reactions in the presence of water (e.g., olivine to serpentine) (Ballentine et al., 2025; Sherwood Lollar et al., 2014). LithoGas provides fast and efficient Monte Carlo modelling of H_2 and ^{4}He generation rates, following the radiolysis quantification methods of Warr et al. (2023) and serpentinization methods of Ardakani et al. (In Press). This implementation provides a rapid pathway to digest a large numbers of lithogeochemical samples, while incorporating their natural variability, to understand their H_2 and/or ^{4}He prospectivity over geological time scales, and to scale generation rates to economically relevant source rock volumes.

Statement of Need

New resource modelling tools are urgently required to explore for ^{4}He and natural H_2, given their emerging and critical role in low-carbon energy generation and in research, medical, and industrial processes (Danabalan et al., 2022; IEA, 2024; Jackson et al., 2024; Sherwood Lollar et al., 2025). LithoGas addresses this need by: 1) calculating H_2 and ^{4}He generation rates via radiolysis and serpentinization from rock geochemical and physical properties; 2) back-projecting these generation rates into deep time to estimate cumulative generation; 3) summarising and plotting results, including novel source-rock-volume-scaling plots focused on

DRAFT

resource estimation metrics. These functions provide the foundation for basin-scale modelling of radiolysis- and serpentinization-dominated H_2 systems, analogous to exploration workflows for conventional hydrocarbon systems.

The Monte Carlo approach (monteProd()) follows the equations of (Warr et al., 2023) and (Ardakani et al., In Press), incorporating truncated normal distributions (controlled by min/max/mean/standard deviation parameters) for sample geochemistry (Fe, U, Th, and K concentrations), physical rock properties (rock density and porosity), and fluid properties (fluid density) (Table 1). Two pathways are available for assigning rock physical properties: user-defined sample-specific distributions, or automatic lookup from the Canadian Rock Physical Property Database (Enkin, 2018) based on the known lithology. The Monte Carlo approach allows for users to incorporate natural variability and uncertainty into the H_2 and ^4He generation rate calculations, which is paramount in early phase exploration. If a deterministic rather than probabilistic model is desired, the standard deviation of any model parameter can be set to zero. H_2 generated via serpentinization can be modelled via two methods (Ardakani et al., In Press) depending on data availability: an iron speciation approach using measured Fe_2O_3 and FeO concentrations (monteSerpFeSpecies()), or a total iron approach using bulk Fe_2O_3T where speciation data are unavailable (monteSerpFeTotal()). Both serpentinization methods use the change in Fe^{3+}/FeT ratio between initial and current states to estimate magnetite (Fe_3O_4) formation and the associated stoichiometric H_2 yield.

Monte Carlo results from multiple samples are summarised using monteSum(), which collapses the full trial distribution to minimum, mean, and maximum generation rates per sample group. Source rock volume scaling plots are generated by monteH2Plot() and monteHePlot(), which calculate per m^3 generation rates across a range of source rock volumes (0.1 to 100 km^3), producing log-log plots that allow direct comparison of H_2 and ^4He prospectivity across samples and lithologies (Fig. 1). These source rock volume scaling plots are a simple yet important development, allowing prospecting for natural H_2 and ^4He at the regional scale based on localised generation rates from abundant lithogeochemical samples. A secondary axis on both plots converts molar generation rates (mol/year for specified source rock volume) to mass rates (kg/year for specified source rock volume) for direct economic interpretation.

Lastly, generation rates produced by monteProd() can be projected into deep time with the deepTimeProd() function. H_2 produced via radiolysis as well as ^4He, U, Th, and K concentrations are back-calculated using radioactive decay law (Fig. 2). For serpentinization, an average rate of serpentinization is applied. These can the then be used by the user to look at cumulative volumes produced over time.

Conclusion

Overall, LithoGas provides a straightforward workflow for modelling multiple samples from a single structured input dataframe through to publication-ready summaries and plots that inform exploration. Functions include: 1) performing the Monte Carlo modelling via monteProd(), ingesting data as a structured dataframe (Table 1); 2) summarising Monte Carlo results via monteSum() and plotting novel source-volume-scaling plots via monteH2Plot() and monteHePlot() (Fig. 1). Example datasets include dataframes structured for Monte Carlo model input with and without known rock properties (structuredDF) and the summarised lithology distributions from The Canadian Rock Physical Property Database (CRPPData).

Table 1: Columns required for different models within monteProd() function. monteProd() gathers input data by column name (e.g., $uMin, $rockDenMax), as such column names must match exactly. Parameters listed with "..." end of column the name (e.g., "u...") require a min, max, mean, and standard deviation columns (e.g., "uMin", "uMax", "uMean", "uSD").

DRAFT

Parameter	Type	Description	Model
Sample	Text	Sample name – not necessarily named "Sample" but some sort of unique sample ID	All models
Age_Ma	Numeric	The age of the rock in millions of year (Ma)	All models
AgeUnc2S_Ma	Numeric	Uncertainty as of age (2S /2σ) in absolute terms (Ma)	All models
u...	Numeric	Minimum, maximum, mean, standard deviation of modelled distribution of uranium concentration - in ppm. Each uMin, uMean, uMax, uSD, are separate columns.	Radiolysis – monteRad()
th...	Numeric	Minimum, maximum, mean, standard deviation of modelled distribution of thorium concentration - in ppm. Each thMin, thMean, thMax, thSD, are separate columns.	Radiolysis – monteRad()
k...	Numeric	Minimum, maximum, mean, standard deviation of modelled distribution of potassium concentration - in weight percent potassium (wt% K). Each kMin, kMean, kMax, kSD, are separate columns.	Radiolysis – monteRad()
fluDen...	Numeric	Minimum, maximum, mean, standard deviation of modelled distribution of fluid density in g/cm3. Each fluDenMin, fluDenMax, fluDenMean, fluDenSD, are separate columns	Radiolysis – monteRad()
Fe3FeTRatInit...	Numeric	Minimum, maximum, mean, standard deviation of modelled distribution of initial Fe^{3+}/FeT ratio at time of rock formation/crystallisation. This ratio is unitless.	Serpentinization, both monteSerpFeSpecie and monteSerpFeTotal(
Fe2O3...	Numeric	Minimum, maximum, mean, standard deviation of modelled distribution of measured concentration of Fe_2O_3, as per iron speciation measurements (iron titration, Mossbauer). In units of weight percent Fe_2O_3 (wt% Fe2O3).	Serpentinization - monteSerpFeSpecie
FeO...	Numeric	Minimum, maximum, mean, standard deviation of modelled distribution of measured concentration of FeO, as per iron speciation measurements (iron titration, Mossbauer). In units of weight percent FeO (wt% FeO).	Serpentinization - monteSerpFeSpecie
Fe2O3T...	Numeric	Minimum, maximum, mean, standard deviation of modelled distribution of measured total iron represented as Fe_2O_3. In units of weight percent Fe_2O_3 (wt% Fe_2O~3'T).	Serpentinization - monteSerpFeTotal(
Fe3FeTRatCur...	Numeric	Minimum, maximum, mean, standard deviation of modelled distribution of initial Fe^{3+}/FeT ratio at time of measurement. This ratio is unitless.	Serpentinization - monteSerpFeTotal(
litLith	Categorical selection	Lithologic groups from Enkin et al. (2018), inclusion of a categorial rock type will draw rock density and porosity from the known distribution and ignore porosity and density distributions input. If the column is left blank, or the column is not included, the model will pull from the rock density and porosity columns	All models, if not assigned by rockDen and por Min/Max/Mean/SD

DRAFT

Parameter	Type	Description	Model
rockDen...	Numeric	Minimum, maximum, mean, standard deviation of modelled distribution, in units of g/cm3	All models, if not assigned by litLith field and therefore joinLitProp function
por...	Numeric	Minimum, maximum, mean, standard deviation of modelled distribution of rock porosity (grain porosity) in fractional porosity (0-1)	Radiolysis, if not assigned by litLith field and therefore joinLitProp function

Figures

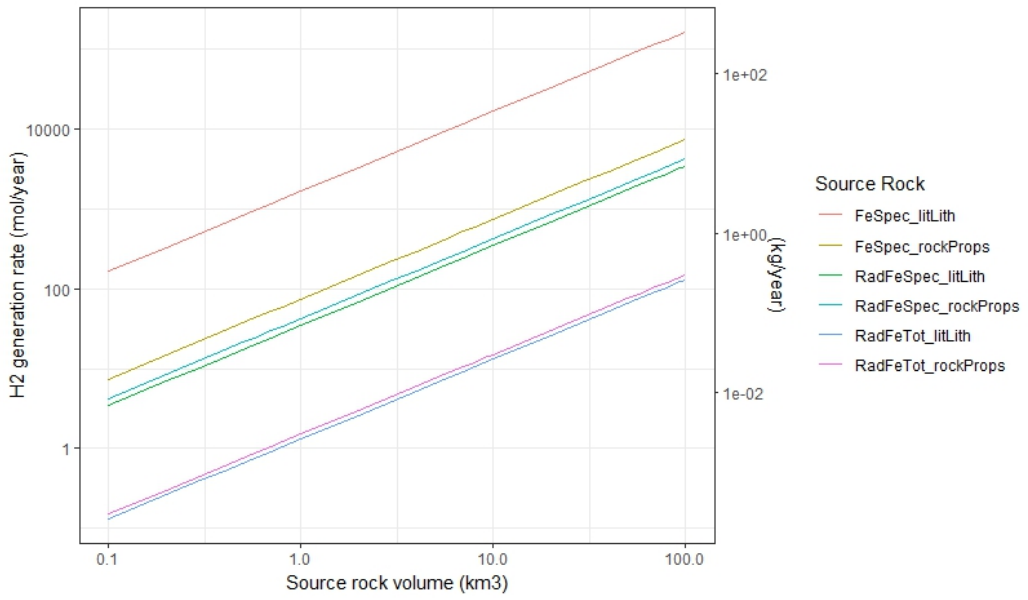


Figure 1: Novel source-rock-volume scaling plots created by LithoGas that used to compare the prospectivity of lithogeochemical samples. The mean generation rate of Monte Carlo distributions are shown scaling from 0.5 km^3 source area to 100 km^3 source area.

DRAFT

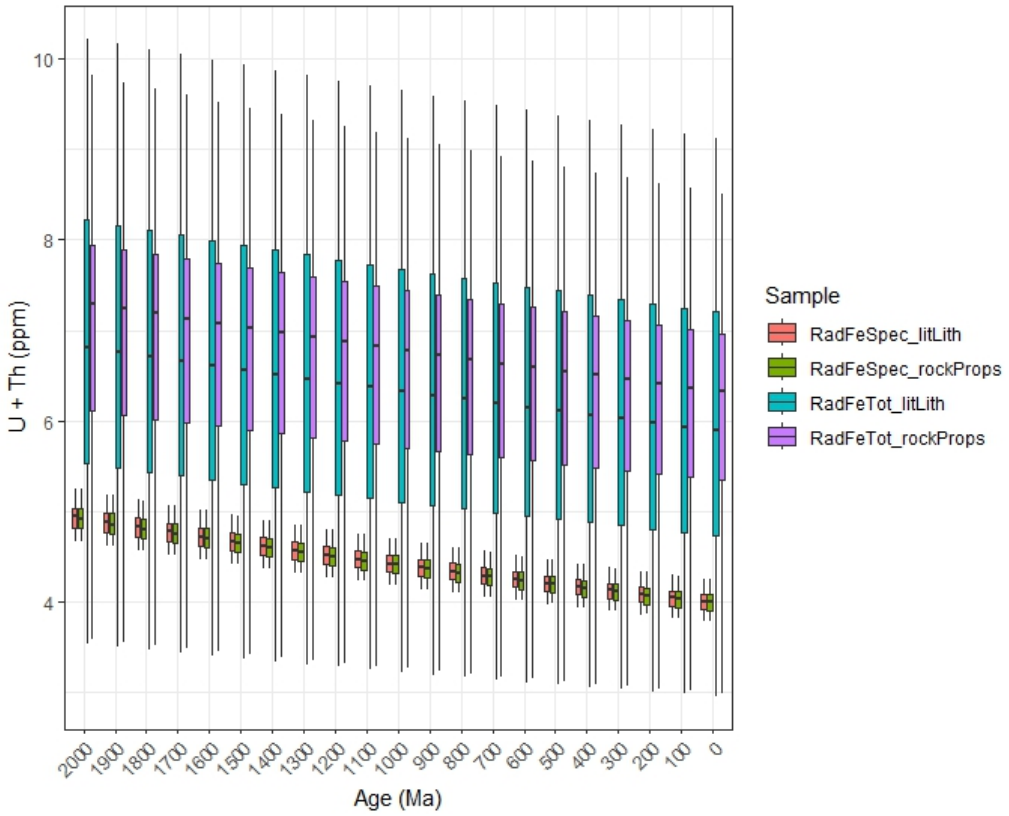


Figure 2: Decay of uranium and thorium over geological time scales, lowering the H_2 and He production rate through radiolysis.

DRA

Anderson, S. T. (2018). Economics, helium, and the u.s. Federal helium reserve: Summary and outlook. *Natural Resources Research*, *27*, 455–477. https://doi.org/https://doi.org/10.1007/s11053-017-9359-y

Ardakani, O. H., Coutts, D. S., Warr, O., Sherwood-Lollar, B., DeBlonde, D., Kabanov, P., & Lister, C. (In Press). A geological framework for natural hydrogen and helium resources in canada. *Earth Science Reviews*. https://doi.org/https://doi.org/10.1016/j.earscirev.2026.105717

Ballentine, C. J., Karolyte, R., Cheng, A., Sherwood Lollar, B., Gluyas, J. G., & Daly, M. C. (2025). Natural hydrogen resource accumulation in the continental crust. *Nature Reviews Earth and Environment*, *6*, 342–356. https://doi.org/10.1038/s43017-025-00670-1

Coveney, R. M., Goebel, E. D., Zeller, D., E. J., & Angio, E. E. (1987). Serpentinization and the origin of hydrogen gas in kansas. *AAPG Bulletin*, *71*, 39–48. https://doi.org/10.1306/94886D3F-1704-11D7-8645000102C1865D

Danabalan, D., Gluyas, J. G., Macpherson, C. G., Abraham-James, T. H., Bluett, J. J., Barry, P. H., & Ballentine, C. J. (2022). The principles of helium exploration. *Petroleum Geoscience*, *28*, 2021–2029. https://doi.org/10.1144/petgeo2021-029

Enkin, R. J. (2018). The canadian rock physical property database: First public release. *Geological Survey of Canada, Open File*, *8460*. https://doi.org/10.4095/313389

Higgins, P. M., Song, M., Warr, O., & Sherwood Lollar, B. (2025). Natural H2 and sulfate production via radiolysis in low porosity and permeability crystalline rocks. *JGR Biogeosciences*, *130*, 1–22. https://doi.org/10.1029/2025JG008863

Hu, Z., & Li, J. (2025). A review of helium resources and development. *Natural Gas Industry B*, *12*, 356–367. https://doi.org/10.1016/j.ngib.2025.05.008

IEA. (2024). *Global hydrogen review 2024*. Online Report. https://www.iea.org/reports/global-hydrogen-review-2024

Jackson, W., Lawrence, S. E., Hutchinson, I. P., Stocks, A. E., Barnicoat, A. C., & Powney. (2024). Natural hydrogen: Sources, systems and exploration plays. *Geoenergy*, *2*. https://doi.org/10.1144/geoenergy2024-002

Johnson, N., Liebreich, D. M., M. Kammen, Ekins, P., McKenna, R., & Staffell, I. (2025). Realistic roles for hydrogen in the future energy transition. *Nature Reviews Clean Technology*, *1*, 351–371. https://doi.org/10.1038/s44359-025-00050-4

Lin, L., Hall, J., Lippmann-Pipke, J., Ward, J. A., Sherwood Lollar, B., DeFlaun, M., Rothmel, R., Moser, D., Gihring, T. M., Mislowack, B., & Onstott, T. C. (2005). Radiolyitc H2 in continental crust: Nuclear power for deep subsurface microbial communities. *Geochemistry, Geophysics, Geosystems*, *6*, 1–13. https://doi.org/10.1029/2004GC000907

Mayhew, L. E., Ellison, E. T., McCollom, T. M., Trainor, T. P., & Templeton, A. S. (2013). Hydrogen generation from low-temperature water–rock reactions. *Nature Geoscience*, *6*, 478–484. https://doi.org/https://doi.org/10.1038/ngeo1825

Miaga, O., Deville, E., Laval, J., Prinzhoefer, A., & Boubacar Diallo, A. (2023). Characterization of the spontaneously recharging natural hydrogen reservoirs of bourakebougou in mali. *Scientific Reports*, *13*. https://doi.org/10.1038/s41598-023-38977-y

Sherwood Lollar, B., Ball, P., Ballentine, C. J., Bordmann, V., Daly, M., Bossart, C. de, Edlmann, K., Gluyas, J., Lacinska, A., Lindsay, J., McIntosh, J., & Truche, L. (2025). *Natural hydrogen: Future energy and resources royal society of london policy briefing*. Online Report. https://royalsociety.org/news-resources/projects/low-carbon-energy-programme/natural-hydrogen/

Sherwood Lollar, B., Onstott, T. C., Lacrampe-Couloume, G., & Ballentine, C. J. (2014). The contribution of the precambrian continental lithosphere to global H2 production. *Nature*, *516*, 379–382. https://doi.org/10.1038/nature14017

Sherwood Lollar, B., & Warr, O. (2026). Decadal record of continental H2 reservoirs reveals potential for subsurface microbial life and natural H2 exploration. *PNAS*, *123*, e2603895123. https://doi.org/https://doi.org/10.1073/pnas.2603895123

Truche, L., Donze, F. C., Goskolli, E., Muceke, B., Loisey, C., Monnin, C., Dutoit, H., & Cerepi, A. (2024). A deep reservoir for hydrogen drives intense degassing in the bulqize ophiolite. *Science*, *383*, 618–621. https://doi.org/10.1126/science.adk9099

Warr, O., Song, M., & Sherwood Lollar, B. (2023). The application of monte carlo modelling to quantify in situ hydrogen and associated element production in the deep subsurface. *Frontiers in Earth Science*, *11*, 1–14. https://doi.org/10.3389/feart.2023.1150740

DRAFT